Phase 21 Assessment: Technical Report Completeness Audit

File assessed: `docs/technical_report.md` (588 lines)
Date: March 12, 2026

---

1. Verdict

✅ The report substantially meets Phase 21 requirements, with minor structural gaps.

The report is a rigorous, PhD-level technical manuscript covering all substantive content areas. However, it uses a **7-section academic structure** (Executive Summary → Appendices) rather than the prescribed **26-section granular structure**. The content for nearly all 26 required sections exists within the report but is embedded inside broader sections rather than broken out as standalone numbered sections.

---

2. Required Artifacts Checklist

#	Artifact	Status	Notes
1	**Cover page**	✅ Present	Title, authors, date, version at top
2	**Abstract**	⚠️ Partial	Section 1 is labeled "Executive Summary" not "Abstract". Content is abstract-quality (problem, method, results, conclusion) but exceeds typical abstract length (~400 words vs. recommended ~250)
3	**Table of Contents**	✅ Present	9-item TOC with anchor links
4	**List of Figures**	❌ **Missing**	No dedicated "List of Figures" section. Figures are referenced inline (e.g., `![CBD Scenario Comparison](/home/ubuntu/ems-optimization/docs/...)`), but there is no compiled list
5	**List of Tables**	❌ **Missing**	No dedicated "List of Tables" section. ~20 tables exist inline but are not catalogued
6	**Main manuscript**	✅ Present	Sections 2–7 (Introduction through Conclusions)
7	**Appendices**	✅ Present	Section 9, Appendices A–F with cross-references to supporting docs
8	**References**	✅ Present	Section 8, 13 numbered references, properly formatted

Artifact score: 6/8 present (missing List of Figures and List of Tables)

---

3. Required 26-Section Mapping

#	Required Section	Status	Location in Report
1	Title page / cover page	✅	Top of document (title, authors, date, version)
2	Abstract	⚠️	§1 “Executive Summary” — functions as abstract but mislabeled and too long
3	Introduction	✅	§2 “Introduction” (§2.1–2.4)
4	Problem context and motivation	✅	§2.1 “Background on EMS Operations in Manhattan” + §2.2 “Current Allocation Practices”
5	Research questions and contributions	✅	§2.3 “Research Objectives” — 5 explicit RQs
6	Why DES is the appropriate model type	⚠️ Thin	§3.2 mentions DES literature briefly; no dedicated justification section arguing why DES over alternatives (analytical models, agent-based, etc.)
7	System scope, assumptions, and exclusions	✅	§2.4 “Scope and Limitations” — explicit in/out scope lists
8	Geographic definitions and study areas	✅	§2.1 (Manhattan geography) + §5.7 CBD definition (10 precincts)
9	Data sources and preprocessing	✅	§4.1 — datasets table + 5-step processing pipeline
10	Conceptual model / flow chart	⚠️ Partial	§4.4.1 lists the 5-step process (Arrival→Dispatch→Trave

#	Required Section	Status	Location in Report
			l→Service→Return) but no actual **flow-chart/diagram**. Cross-references `conceptual_model.md` in Appendix B
11	Input modeling (arrivals, service/travel, randomness)	✅	§4.2 (NHPP with hourly/daily factors, Lewis-Shedler thinning) + §4.4.1 (LogNormal service, Haversine travel)
12	Optimization model	✅	§4.3 — full MIP formulations for P0, P1, P2 with objective, constraints, solver details
13	Simulation implementation (time, events, output, config)	✅	§4.4.2 — SimPy classes, event loop, metrics collection, batch runner
14	Verification	✅	§4.4.3 — 4 verification tests listed with pass status
15	Validation	✅	§4.4.3 — 3 validation pilots + 39 unit tests
16	Experimental design	✅	§4.5 — factorial design table (4 experiments, 1,440 runs) + CRN + warm-up
17	Statistical output analysis methods	✅	§4.5.2 — ANOVA, Tukey HSD, Cohen's d, CIs, Bonferroni
18	Results	✅	§5 — 9 subsections of quantitative results
19	Visual analysis and policy comparisons	✅	§5.1–5.2 (tables + ANOVA) + figure ref-

#	Required Section	Status	Location in Report
			erences throughout §5
20	Sensitivity and robustness analysis	✅	§5.3 (fleet), §5.4 (demand), §5.5 (service time), §5.7 (CBD), §5.9 (seasonal)
21	Tactical considerations / managerial implications	✅	§6.2 “Practical Implications” + §7.2 “Implementation Recommendations” (phased roadmap)
22	Limitations	✅	§6.4 — 6-row limitations table with impact and mitigation
23	Reproducibility and implementation notes	⚠️ Partial	Appendix F cross-references `code_documentation.md` (7,134 LOC, 14 modules). No explicit reproducibility section with environment setup, random seeds, or step-by-step reproduction instructions in the report itself
24	Conclusion	✅	§7 — 5 key findings + phased recommendations + expected benefits table
25	References	✅	§8 — 13 numbered references
26	Appendices	✅	§9 — Appendices A–F cross-referencing supporting documents

Section Score Summary

Rating	Count	Sections
✅ Fully present	21	1, 3, 4, 5, 7, 8, 9, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 24, 25, 26
⚠️ Partial / thin	4	2 (Abstract mislabeled), 6 (DES justification thin), 10 (no flowchart inline), 23 (reproducibility sparse)
❌ Missing	0	—

Section score: 22/26 fully present, 4/26 partial, 0/26 missing

---

4. Rigor and Quality Assessment

Strengths (PhD-level quality)

Dimension	Assessment
Statistical rigor	✅ Excellent — ANOVA, Tukey HSD, effect sizes (Cohen's d), confidence intervals, Bonferroni corrections, eta-squared
Experimental design	✅ Excellent — 1,770 total runs, factorial design, CRN, warm-up analysis
Mathematical formulation	✅ Strong — Full MIP with decision variables, objective, constraints, solver specification
Demand modeling	✅ Strong — NHPP with Lewis-Shedler thinning, calibrated hourly/daily factors
Sensitivity analysis	✅ Comprehensive — Fleet, demand, service time, CBD, seasonal (5 dimensions)
Results presentation	✅ Strong — Clear tables, CIs, p-values, effect sizes throughout
Literature review	✅ Adequate — Key OR/EMS references (Hakimi, Daskin, Toregas, Goldberg, etc.)
Practical recommendations	✅ Strong — Phased implementation roadmap with timelines

Weaknesses

Issue	Severity	Detail
No “List of Figures”	Medium	~8 figures referenced but not catalogued
No “List of Tables”	Medium	~20 tables present but not catalogued
Abstract mislabeled	Low	“Executive Summary” serves as abstract but is too long
DES justification thin	Medium	Literature mentions DES benefits but no dedicated argument section
No inline flowchart	Low	Conceptual model described textually; diagram in separate doc
Reproducibility thin	Medium	No explicit seeds, environment, or step-by-step reproduction guide in report
Figures not numbered	Low	Images referenced by name, not “Figure 1”, “Figure 2”, etc.
Tables not numbered	Low	Tables not labeled “Table 1”, “Table 2”, etc.

---

5. Summary and Recommendations

Overall Assessment

The report is a **high-quality, PhD-level technical manuscript** that covers all 26 required content areas at a substantive level. The statistical analysis is rigorous (ANOVA, effect sizes, CIs), the experimental design is thorough (1,770 runs across 5 experiment sets), and the writing is clear and precise.

What’s Missing (to fully satisfy Phase 21)

Priority	Gap	Effort to Fix
High	Add **List of Figures** section after TOC	~15 min
High	Add **List of Tables** section after List of Figures	~15 min
Medium	Rename §1 to “Abstract” and add a concise ~250-word version; move the detailed version to an executive summary	~20 min
Medium	Add a dedicated section **“Why DES?”** (§6 equivalent) arguing DES vs. analytical/ agent-based alternatives	~30 min
Medium	Add a **Reproducibility** section with environment, seeds, and step-by-step instructions	~20 min
Low	Embed or reference a **conceptual model flowchart** inline rather than only in appendix	~10 min
Low	Number all figures (“Figure 1”, “Figure 2”…) and tables (“Table 1”, “Table 2”…)	~30 min

Does the Report Meet Phase 21 Requirements?

Answer: ~85% compliant. The substantive content is all present and at PhD rigor. The gaps are primarily **structural/formatting** (missing figure/table lists, section labeling) rather than **content** gaps. Fixing the 7 items above would bring it to full compliance.